1.4 Perimeter and Area in the Coordinate Plane

Name _____ Period ____ Date _____

Geometry 1.4 · Student sheet

I can…

- classify a polygon by its number of sides and as convex or concave.
- find the perimeter of a polygon in the coordinate plane.
- find the area of a polygon in the coordinate plane.

Vocabulary · fill in the missing word

Polygon — a closed plane figure formed by three or more segments that meet only at their _____

n-gon — a _____ with n sides

Convex polygon — a polygon in which no line that _____ a side passes through the interior

Concave polygon — a polygon that is not _____

Part 1 · Discovery: around and inside on a grid

Plot A(−2, 1), B(4, 1), and C(−2, 9).

AB	AC	BC	Perimeter	Area

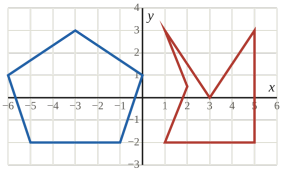
Which side needed the Distance Formula? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Classifying polygons: name a polygon by its number of sides; it is convex if no line containing a side passes through the _____.

Example 1

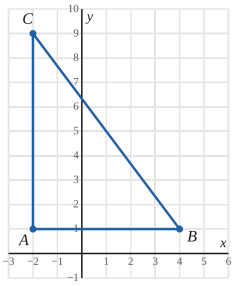
Classify each polygon by its number of sides. Tell whether it is convex or concave.



Perimeter on the coordinate plane: count horizontal and vertical sides; use the _____ for each slanted side; then add.

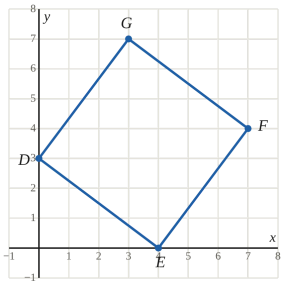
Example 2

Find the perimeter of △ABC with A(−2, 1), B(4, 1), and C(−2, 9).



Example 3

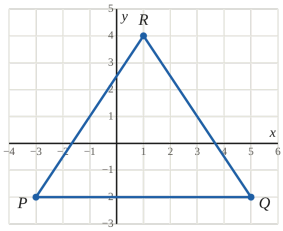
Find the perimeter and area of the quadrilateral with vertices D(0, 3), E(4, 0), F(7, 4), and G(3, 7).



Area: for a triangle, A = ½bh, where the height is _____ to the base.

Example 4

Find the area of △PQR with P(−3, −2), Q(5, −2), and R(1, 4).



Example 5

A garden has corners at (0, 0), (12, 0), (12, 9), and (0, 9), in feet. Fencing costs $5 per foot. Find the cost of the fence and the area of the garden.

Part 3 · Partner practice

Partner 1

Find the perimeter of the triangle with vertices (1, 1), (1, 5), and (4, 1).

Partner 2

Find the area of the rectangle with vertices (−2, −1), (4, −1), (4, 3), and (−2, 3).